Analog to Digital Conversion: Quantization and Coding

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Key Concepts

1) Digital computers represent numbers using a finite number of bits, which implies there are a finite number of amplitude levels possible.

2) Conversion of an analog signal to digital format is performed by a device called an analog-to-digital (A/D) converter.

    a) The A/D converter samples the signal amplitude at discrete instants in time.

    b) The A/D quantizes or rounds the amplitude of the signal to a discrete set of amplitudes.

c) The quantized amplitude is represented by a binary code.

3) Quantizers are usually assumed to be memoryless and time invariant.

   a) Memoryless: the quantized value depends only on the current amplitude of the signal.

   b) Time invariant: the quantization does not change over time.

4) A uniform quantizer has a constant step size between allowable signal levels.

5) Quantization error is the difference between the original and quantized values of the signal.

6) The absolute value of the quantization error is bounded by one-half the quantization step size.

7) if R is the range (the maximum quantized level minus the minimum) of the A/D, then the step size or resolution is R divided by the number of levels.

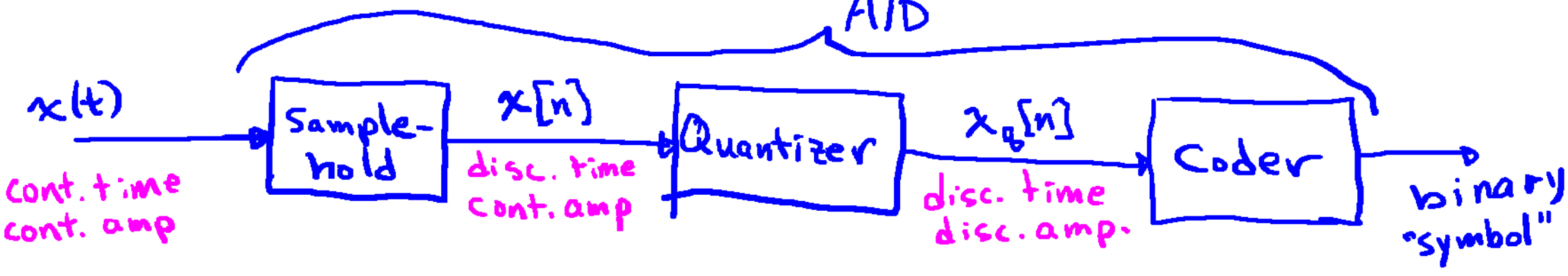
8) Coding is the process of assigning a binary number to each quantization level.

Video Screenshots

A/D Conversion - Quantization + Coding

Realistic A/D: limited amplitude range
limited set of possible values

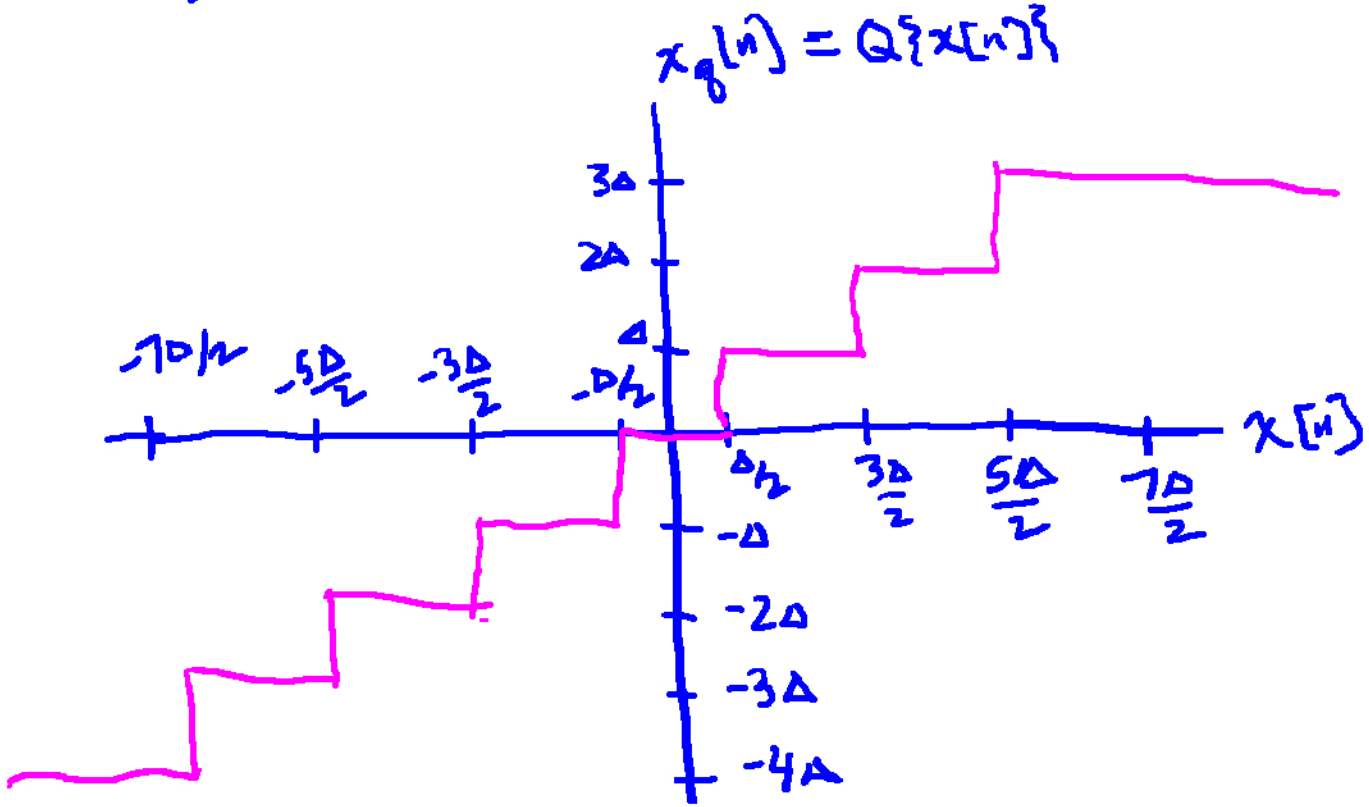
Example: digital camera A/D has 14 bits ⇒
2^{14} possible amplitudes at each pixel



$$x_q[n] = Q\{x[n]\}$$

Quantizers $Q\{\cdot\}$

1) Memoryless – $x_q[n]$ depends only on $x[n]$
2) Time-invariant – $Q\{\cdot\}$ doesnot change with n
3) Uniform – constant step size between levels (Δ)

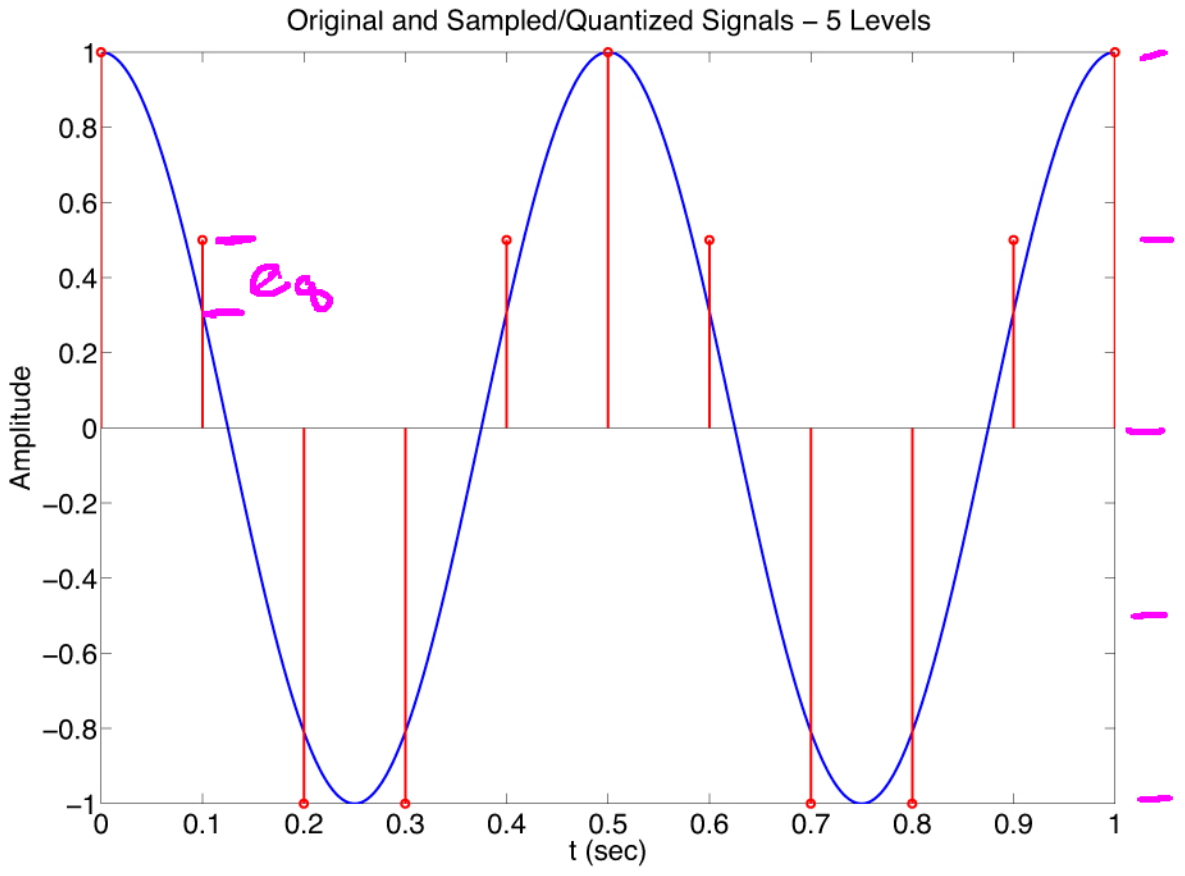


1) Overflow

2) Quant. error

$$e_q[n] = x_q[n] - x[n]$$

$$-\frac{\Delta}{2} \le e_q[n] \le \frac{\Delta}{2}$$


Original and Sampled/Quantized Signals – 5 Levels
Amplitude
t (sec)
1
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
−1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

T = 0.1 sec

Coding

Assign a binary number to each quant. level

Examples: 2's complement, sign/magnitude, etc

Given b+1 bits: up to 2^{b+1} quant. levels

If R is range (max - min):

Resolution (step size) $\Delta = \frac{R}{2^{b+1}}$

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